

PRINCE CHAUHAN

SENIOR QUANTITATIVE ANALYST | SYSTEMATIC DERIVATIVES & STATISTICAL ARBITRAGE

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SEBI Registered Research Analyst

PROFESSIONAL SUMMARY

Senior Quantitative Analyst and Systematic Derivatives Strategist with **9+ years of experience** designing, validating, and trading systematic options volatility and statistical arbitrage strategies across US (SPX/SPXW, CBOE VIX) and Indian (NSE/BSE) equity index derivatives via Interactive Brokers (IBKR) and domestic broker gateways. Specialized in implied volatility surface modeling (SVI, cubic splines), cointegration, unsupervised market regime detection, and microstructure-aware execution engines with strict risk and capital constraints. Combines production-grade Python research pipelines (Polars, NumPy, SciPy, Asyncio) with advanced financial accounting forensics and regulatory qualifications (CA Final Group 1, B.Com Hons 1st Div, SEBI Registered Research Analyst).

PROFESSIONAL EXPERIENCE

SMC Global Securities Ltd. - Senior Quantitative Analyst, Systematic Strategies & Quant Desk Dec 2024 - Present

- Architected causal, event-driven simulation pipelines and point-in-time alpha research infrastructure on tick-level and 1-minute intraday bar feeds for US and Indian index derivatives (SPX, CBOE VIX, Nifty, Bank Nifty), eliminating lookahead bias in intraday studies.
- Engineered automated execution pipelines and volatility surface modeling for US equity index options (SPX/SPXW 0DTE) via Interactive Brokers (IBKR API), systematically capturing short-horizon IV skew mispricings while managing dynamic intraday delta rebalancing.
- Developed Bank Nifty basket statistical arbitrage models applying Johansen cointegration rank tests, dynamic rolling beta hedging, and Ornstein-Uhlenbeck spread half-life modeling across heavyweight banking equities to capture mean-reversion opportunities.
- Calibrated systematic options strategies delivering 2.5+ Out-of-Sample Sharpe and 2.1+ Profit Factor in strictly chronological simulations after pre-deducting conservative 0.50% round-trip execution friction (bid-ask spread crossing, market impact, statutory costs).
- Transitioned legacy rules-based execution to an asynchronous event-driven tick pipeline (Python Asyncio, broker WebSockets, memory-mapped IPC), improving fill efficiency by 24% and minimizing bid-ask crossing drag and execution slippage across multi-leg orders.
- Managed with proprietary risk desks to implement real-time portfolio Greeks monitoring (Delta, Gamma, Vega, Theta), dealer Gamma Exposure (GEX) mapping, parametric/historical VaR, automated volatility kill switches, and SPAN cross-margin capital limits.

Smart Bulls Capital - Research Analyst, Derivatives & Quantitative Research Mar 2017 - Nov 2024

- Researched, structured, and traded systematic multi-leg options volatility strategies (calendar spreads, ratio butterflies, iron condors) with dynamic delta and vega rebalancing using real-time IV surface spline smoothing.
- Built Black-Scholes-Merton PDE inversion models (Newton-Raphson, Brent solvers), cubic spline & SVI strike smoothing, and real-time dealer Gamma Exposure (GEX) profiles to map institutional pin risk.
- Conducted fundamental equity research including financial statement forensics, discounted cash flow (DCF) modeling, and valuation across banking, energy, and industrial sectors to inform desk capital deployment to quant strategies.
- Managed daily P&L attribution, margin utilization, order slice execution, and drawdown limits across multi-strike options books, ensuring strict adherence to risk budgets and desk constraints.

Earlier Corporate Experience - Corporate Finance & Audit Roles 2014 - 2017

- Managed treasury operations, working capital forecasting, and forensic variance audits at Indian Oil Corporation Ltd. (IOCL, 2016 - 2017); performed statutory audits and corporate tax structuring at CA Matla & Associates (2014 - 2016).

QUANTITATIVE RESEARCH & ARCHITECTURE

- Point-in-Time Alpha Simulation Framework & Audit Gate:** Designed modular Python framework utilizing 7+ years of causal 1-minute historical data with zero lookahead bias; pre-deducts 0.50% turnover friction, enforces clearing-house SPAN margin models, portfolio leverage caps, point-in-time invariant audit framework, 70/20/10 Walk-Forward Optimization, and 1,000-path Monte Carlo drawdown cones.
- Arbitrage-Free Options Volatility Surface & GEX Profiling:** Formulated Black-Scholes inversion using Newton-Raphson/Brent solvers, cubic spline & SVI surface fitting, arbitrage-free total variance constraints, and dealer Gamma Exposure (GEX) sensitivity profiling to stabilize IV surfaces and GEX maps across strikes and tenors.
- Bank Nifty Basket Cointegration & Statistical Arbitrage:** Developed Johansen multivariate cointegration rank tests, dynamic rolling beta weighting, Ornstein-Uhlenbeck spread half-life estimation, automated strike re-centering thresholds, and contract-rollover rules while maintaining strict delta neutrality.
- Market Regime Volatility Gating & Microstructure Simulator:** Engineered unsupervised K-Means clustering on Parkinson realized volatility, IV/RV ratios, and return skewness, reducing simulated maximum drawdown by 28%; paired with an execution simulator modeling order book queue priority, non-linear market impact, and fill slippage.

TECHNICAL & QUANTITATIVE COMPETENCIES

- Quantitative Research:** Statistical Arbitrage, Johansen Cointegration, Dynamic Beta Hedging, SVI Volatility Surface Fitting, Implied vs Realized Volatility (IV/RV, Parkinson), Market Regime Detection (K-Means), Walk-Forward Optimization (WFO), Monte Carlo Simulation.
- Derivatives & Risk Governance:** Options Pricing (Black-Scholes), Portfolio Greeks (Delta, Gamma, Vega, Theta), Delta/Vega Neutrality, Dealer Gamma Exposure (GEX), Parametric & Historical VaR, SPAN Cross-Margin Optimization, Portfolio Leverage Constraints.
- Programming & Infrastructure:** Python 3.11+, Polars, DuckDB, NumPy, SciPy, Statsmodels, Scikit-Learn, VectorBT, SQL, Apache Parquet, Asyncio, WebSockets, Memory-Mapped IPC (mmap), Interactive Brokers API (IBKR / ib_insync), ReportLab PDF Engine, Git, Linux/Bash.
- Markets & Microstructure:** US & Indian Equity Derivatives (SPX/SPXW 0DTE, CBOE VIX, Nifty, Bank Nifty, Sensex), Single-Stock Futures & Options, Limit Order Book Dynamics, Smart Limit Slicing, Fill Slippage Minimization.

EDUCATION & REGULATORY CREDENTIALS

- Bachelor of Commerce (Honours):** Kirori Mal College, University of Delhi (2011 - 2014) | 71% Marks (First Division)
- Chartered Accountancy (CA Final):** Institute of Chartered Accountants of India (ICAI, 2018) | 78% Marks (Cleared Group 1)
- Regulatory Licenses:** SEBI Registered Research Analyst | NISM Series-XV: Research Analyst